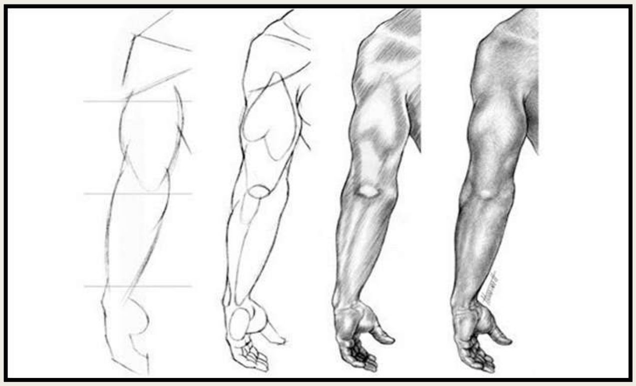
OSTEOPATHIC APPROACH TO THE UPPER EXTREMITY: THE INTEGRATED UPPER EXTREMITY EXAM & CONSIDERATIONS FOR TREATMENT



Dr Sean Moloney

Special thanks to
Dr. Pierce-Talsma

I use the term integration to point out that as osteopath's we look beyond the traditional approach to history taking exams, and treatment options.
In truth, as an osteopath I would not consider this an integrated approach to a shoulder exam I would just consider it a normal approach to the upper extremity.
An osteopathic exam includes all the traditional orthopedic tests.
It is perhaps a subtle distinction, however it is important because of your listening to this lecture you chose to be an osteopath
and that means you should have a broader view than the traditional medical model.

OBJECTIVES

1. Describe and perform upper extremity **orthopedic tests**, correctly interpret findings and identify positive and negative test findings
2. Integrate the standard orthopedic exam with osteopathic structural exam findings
3. Identify somatic dysfunction which can contribute to dysfunction and pain in the upper extremity
4. Discuss **indications/contraindications** and proper application of OMM in the treatment of upper extremity injuries and pain
5. Describe the **functional anatomy and motions permitted (both major and minor)** to evaluate and treat the Upper Extremity, including the shoulder, elbow, forearm, wrist and hand including:

– **Scapulothoracic, Sternoclavicular, Acromioclavicular, Glenohumeral, Radiohumeral, Ulnohumeral, Proximal radioulnar, Distal Radioulnar, Radiocarpal, Intercarpal, Carpometacarpal, Metacarphophalangeal, Interphalangeal**

This lecture is part review (doctoring/ COAR UE) and part new- specifically around the glides and slides (minor motions permitted) for the upper extremity

OMT for the Upper Extremity Helps With…

- Promotion of energy efficient and painless movement
- Optimized balance around the Joints (decreases wear/tear)
- Decreased Stress on Somatic Structures in the Upper Extremity
- Balance of Neural input- Motor/ sensory/ proprioceptive/ autonomic
- Decreased nociceptive drive
- Safe and effective venous and lymphatic return

Integrated Orthopedic Exam and Osteopathic Upper Extremity Exam

- Begin with observing What is their posture? Head forward? Shoulders forward?, IR or ER of the arms? Increased thoracic kyphosis? Scoliosis? One shoulder higher than another? Contours of shoulders, carrying angles, symmetry in shoulders/elbows/wrists
- Inspect Erythema, Edema, Displacement, Warmth, muscle atrophy, increased varas or valgus (elbows wrist), positioning at rest,- compare side to side
- Vasculature/Lymphatic- pulses and lymph nodes
- Neuro Exam- (spinal level innervation and peripheral- strength, sensation, reflexes)
- Palpate/ TART the area of the complaint- and the joint above and below- know what you are palpating (ANATOMY!! Bones, Joints, Ligaments, Tendons, Muscles, Fascia, bursa)
- Active ROM, then Passive ROM, Muscle Strength Testing (Joint above and below, distinguish END FEEL! Grossly lax, restricted or somatic dysfunction. Glides/Slides of Major and Minor Motion)
- Special tests as necessary (test the normal side first)
- Observe for somatic dysfunction of the UE (articular, soft tissue, myofascial, tenderpoints, trigger points etc) as well as the cervical spine, thoracic inlet, upper thorax and ribs as well as T10-L2 (diaphragm attachments & latissimus)
- Observe for somatic dysfunction contributing to the ABC's
  - *Autonomics- T1-T4, Biomechanics- Major/minor motions, glides/slides, articular, muscular/fascial restrictions, Circulation- thoracic inlet, pectoral region, diaphragm, Screening- Mitchell Model, Zink, Upper Crossed, Mood*

You have done much of this before- in your COAR UE
The new piece is adding in the glides and slides (which are the major and minor motions of each joint)

COAR UE EXAM- now we are adding in a new piece- looking for specific somatic dysfunction of the upper extremity.

Anterior & Posterior Seated
Head, neck shoulder
Observe: posture, lateral plumbline, swelling, deformity, skin lesions
Observe: upper crossed considerations (protracted/elevated scapula, IR of shoulders, anterior head, increased thoracic kyphosis, increased cervical lordosis)
Palpate for TART:

Bony structures: acromion, sternoclavicular joint, spine of the scapula, ILA of scapula

Muscular structures: pectoralis major/minor, deltoid, biceps, supraspinatus, infraspinatus, teres major/minor, latissimus, serratus anterior, trapezius

Thoracic spine & ribs

Type I or II dysfunction, viscerosomatic reflexes, physiologic or non

physiologic ribs
The joint above and below
Assess ROM in both active and passive
Shoulder: Flexion, extension, abduction, adduction, internal rotation, external rotation, circumduction
Cervical spine: flexion, extension, sidebending, rotation
Scapula: protraction, retraction, elevation, depression, scapular rotation (observed through arm abduction)
Elbow: flexion, extension
Assess muscular strength around the joint in all ROM and joints above and below (C-spine & elbow)
Special Tests
Neer/Hawkins
Apley Scratch test
Drop arm, empty can
Yearguson's, speeds
Apprehension, sulcus sign
Crossover test (aka scarf sign)
Lift off sign
O'Brien's Active Compression Test

<u>Vascular</u>

Ulnar and radial pulses

<u>Neuro</u>

Observe: muscle bulk and tone
Deep Tendon Reflexes: upper extremities, myotome strength upper extremities, dermatomes, hoffman's
Spurling's maneuver

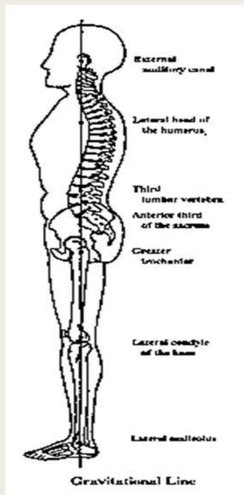
<u>Osteopathic Structural Exam</u>

A- Autonomics- T1-T4
B- Biomechanics- Range of motion
C- Circulation- Thoracic inlet, diaphragm
S- Screening- Zink, posture (done above)

The Integrated Exam Begins with the History

- Ask the usual questions- When did it begin? Any trauma or inciting incident? Is it getting worse? Is it constant or intermittent? Quality of the pain? Pain scale? Radiation of the pain? Exacerbating or alleviating factors? Medications tried? Therapies tried? History of this problem before? Associated symptoms? Specific times of day or aggravating positions or activities? What has been done for the pain so far? One specific injury, or gradually over time? Participation in physical activity?
- Limitation in ROM
- Inability to perform ADL
  - *Reaching overhead, combing hair, fastening bra or tucking in shirt, sleeping on side, lifting*
- Ask about any previous trauma history (no matter what area)
- Any history of imaging, xrays, MRI, etc
- History of PT- and what was done at PT
- Change in physical activity
- Change in job, or other ergonomic factor, chronic posture that they are in at work or at home
- Stress/anxiety/depression
- Recent changes in medical status (health)
- Family history: arthritis, rheumatoid, gout, autoimmune
- Rule out Referral pain with questions (cardiac, GI, diaphragm irritation, neurogenic)

Begin by Observing Posture



- <u>Gravitational Line</u>
  - *External Auditory Meatus*
  - *Lateral head of the humerus*
  - *Body of L3*
  - *Anterior third of the sacrum*
  - *Greater Trochanter*
  - *Lateral condyle of the knee*
  - *Lateral malleolus*

S- Screening

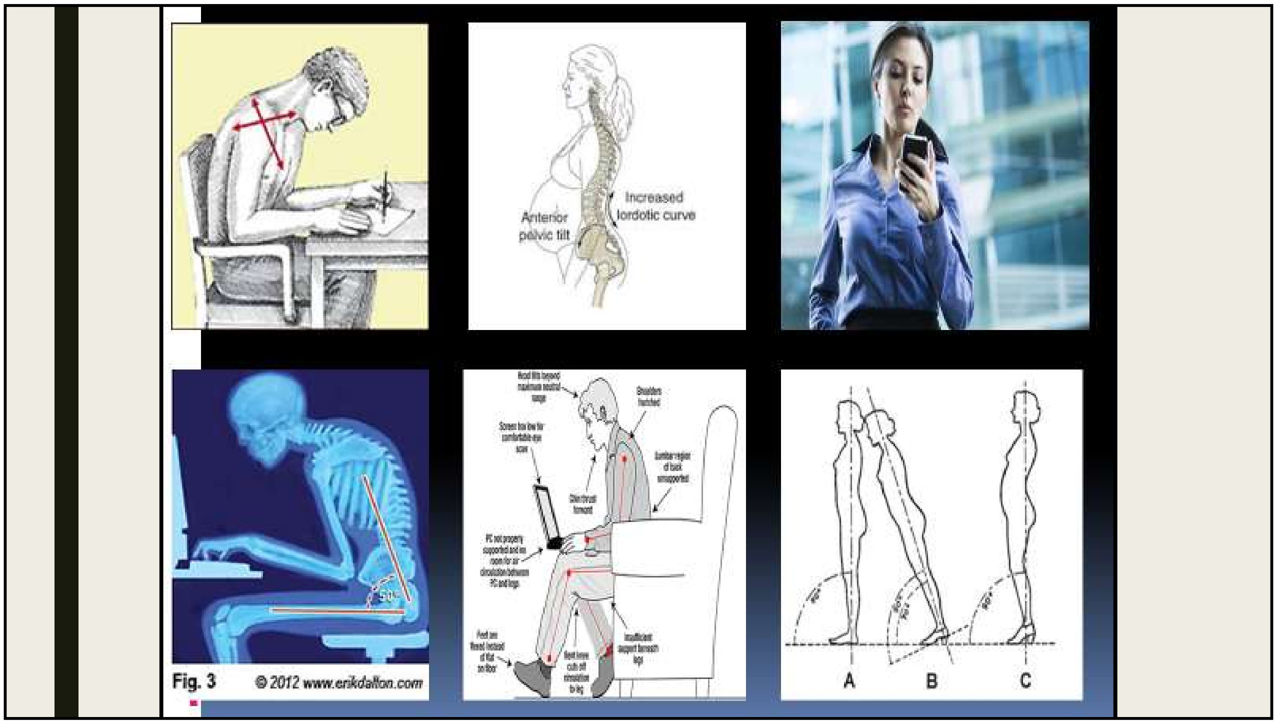
- □ Compare the shoulder heights for symmetry and observe the spine for any kyphosis or scoliosis
- □ Compare the contours of shoulders and shoulder girdle
- □ Compare the carrying angle of the upper extremities
- □ Compare the elbow and wrists for symmetry
- □ Inspect the dorsal surfaces of the extremities for muscle atrophy or wasting, or skin lesions
- □ Inspect the palmar surface of the hands for atrophy of the thenar and hypothenar eminence
- ❑ Upper crossed syndrome picture
- ❑ Leg length discrepancy
- ❑ Mind, Body, Spirit Considerations
- ❑ Ergonomics

FIG. 30-69 Inspection of overall body posture. **A,** Anterior view. **B,** Posterior view. **C,** Lateral view. (From Seidel HM et al: *Mosby's guide to physical examination*, ed 7, St Louis, 2011, Mosby.)

Leg length discrepancy- or sacral base unleveling can cause compensatory postures and altered biomechanics all the way up to the thorax, ribs, cervical spine head and even upper extremity

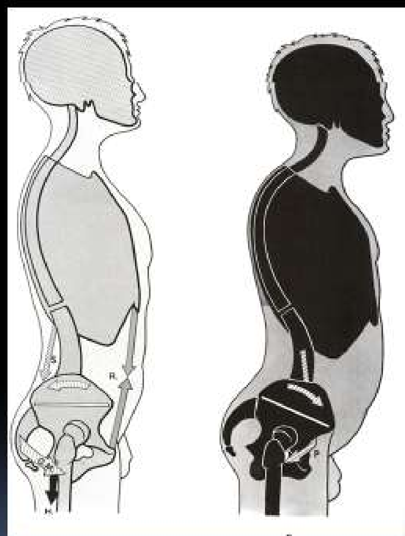
Mind/Body/Spirit Considerations- life stressors can contribute to increase in muscle tensions in the neck and shoulder region

Ergonomics- how does this patient sit, stand, work, sleep

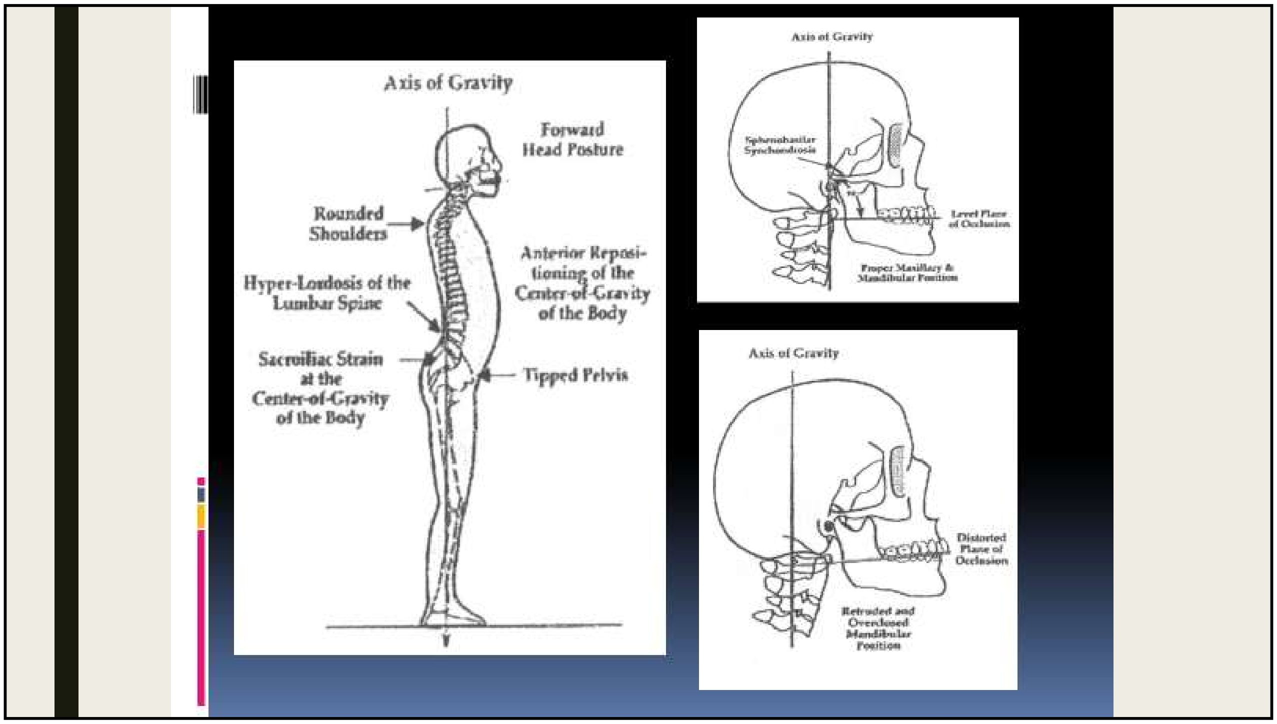


I phone hunch- can add pounds of weight on your neck each time you walk down.

Look at this posture- How is this going to affect:



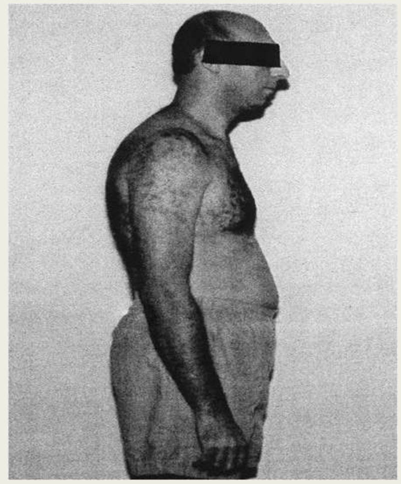
- Respiration
- The Thoracic Inlet- Lymphatics
- Thoracic spine
  - Facilitated spinal segments?
- Shoulder motion



Think about upper crossed in patients with TMJ and headache complaints
Long dural strains from tipped pelvis
But also malalignment of bite

Postural Changes that Affect Muscle Imbalance

- <u>Upper Crossed</u>
  - *Forward Head Posture*
  - *Extension of the OA and upper C-spine*
  - *Increased kyphosis of the C/T junction (Hump at base of the neck)*
  - *Protracted, elevated and internally rotated shoulders*



Short tight LS upper trapz and pectorals
Weak inhibited stabilizers of the scapula rhomboids serratus anterior middle and lower trapezius deep neck flexors

Elevation and protraction of the shoulders
Winged scapula
Forward head posture

CONSIDER THAT RESTRICTIONS IN ONE JOINT WILL LEAD TO COMPENSATORY MOVEMENT IN OTHER JOINTS. ALL JOINTS OF THE SHOULDER NEED TO MOVE FREELY IN ORDER FOR PROPER FUNCTIONING OF THE SHOULDER

"Some factors that predispose the patient to shoulder pain include unstable glenohumeral joints, weakness of scapular stabilizers, abnormal posture, and hypomobility of the cervical and thoracic spine" (Pain Medicine & Managment 2nd edition)

Autonomics & Innervation

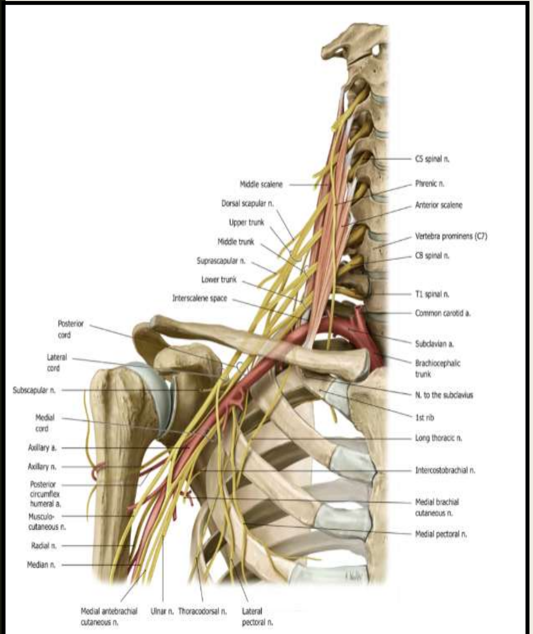
- *Sympathetics*
  - Originate at T1-T4- innervation then goes to the cervical ganglia before making its way to the upper extremities
- *Parasympathetics*
  - Few in the extremities
- *Other Innervation*
  - Brachial plexus
  - C5-T1

Places where the Nerves in the brachial plexus can get hung up

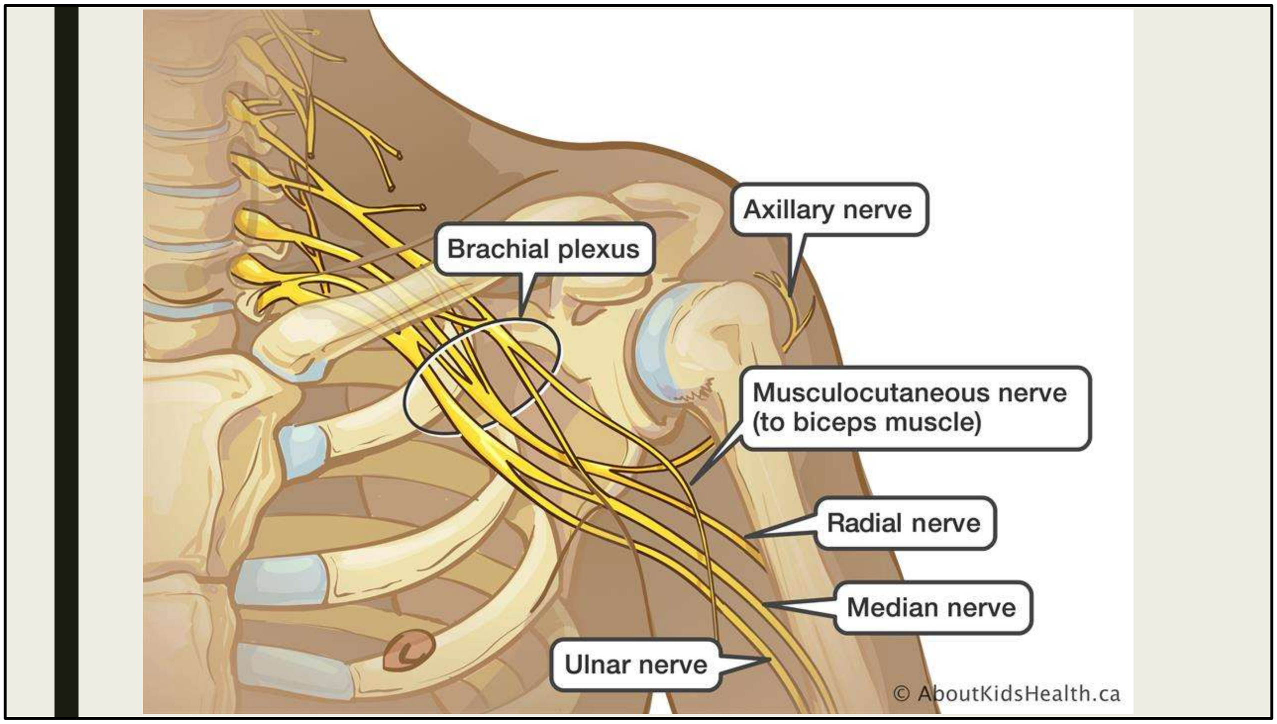
- Nerve roots at cervical foramina
- Between the middle and anterior scalene
- Clavicle and first rib
- Pectoralis minor

Peripheral nerve entrapment

- Median, Radial, Ulnar



COAR UE Review


Axillary nerve
Brachial plexus
Musculocutaneous nerve
(to biceps muscle)
Radial nerve
Median nerve
Ulnar nerve
© AboutKidsHealth.ca

Biomechanics

– *Sternoclavicular joint*

- Important connection of the upper extremity to the thorax- the only true bony joint to do so. Restriction of motion here will lead to restriction of motion at the AC joint.

– *Acromioclavicular joint*

- Connects the scapula to the clavicle and motion is necessary for shoulder abduction to occur.

– *Glenohumeral joint*

- Lots of mobility, little stability. The rotator cuff muscles work to keep it in place and stabilized. Alteration in rotator cuff function or scapular kinematics can affect this joint.
- Commonly the posterior capsule is tight- this causes anterior translation and elevation of the humeral head which could lead to impingement syndromes.

– *Scapulothoracic joint*

- The scapula on the thorax- pseudoarticulation- attached by rhomboids, trapezius, serratus anterior, pectoralis minor and levator scapula. Tension on the muscles controls the position. Changes in the position of the scapula alter the length tension relationships and positions of the rotator cuff muscles and joint arthrokinematics.

– *Elbow*

- Connections from the shoulder (biceps, triceps) have long connections to the elbow. Elbow biomechanics may play a role in shoulder biomechanics.

– *Thoracic spine/Ribs*

- The scapula must glide over the thoracic spine. Somatic dysfunction of the thoracic spine may alter the position of the scapula or affect musculature connecting to the scapula.

– *Cervical spine*

- Many of the muscles that control scapular motion attach or originate from the cervical spine, therefore these areas must be assessed for somatic dysfunction.

– *Pelvis*

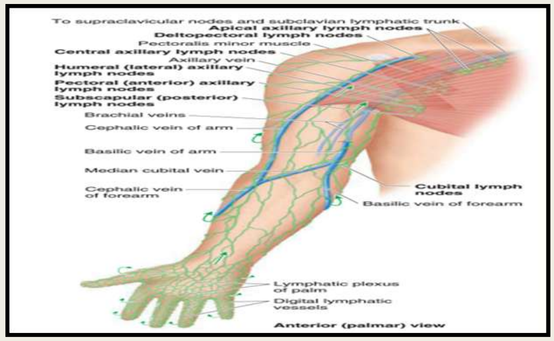
- Don’t miss long musculoskeletal and fascial connections like the latissimus dorsi which attaches onto the iliac crest and the humerus

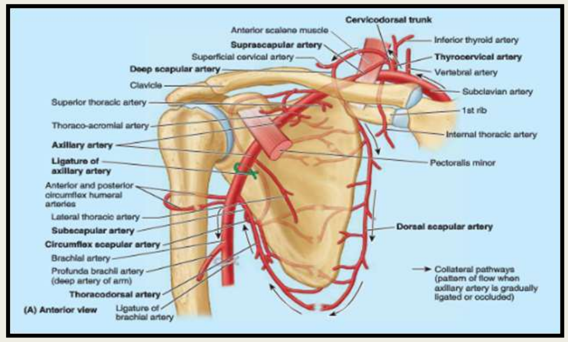
REVIEW FROM COAR UE

Know. This. Slide. Just know it. So many great considerations.

Circulation

- *Many UE complaints are "congestion issues"*
- *Axillary region*- lymphatics from the upper extremity flow to the axillary region before moving into the thoracic duct or subclavian trunk. Myofascial restrictions in this region, including from pectoralis major/minor may impede lymphatic flow.
- *Thoracic inlet*- lymphatic and vascular drainage as well as arterial flow to/ from the arm may be impeded here
- *Diaphragm*- major lymph/venous pump
- ■ *Subclavian Artery*- The main artery to the upper extremity. This artery passes through several structures which may impede flow-
- *Anterior and middle scalenes*
- *First rib and clavicle*
- *Pectoralis minor muscle (pectoral fascia)*
- *Fascia of the upper extremity*





I won't test on artery, lymph, vein vessel names but you should know subclavian artery and were it can become impeded and that the lymph goes into the axillary region, under the pectoralis

The Neuro Exam

- <u>Reflexes</u>
  - *Biceps C5*
  - *Brachioradialis C6*
  - *Triceps C7*
- <u>Dermatomes</u>- see picture
- <u>Myotomes</u>
  - *Shoulder Shrug- C5*
  - *Elbow Flexion- C6*
  - *Elbow Extension- C7*
  - *Wrist Flexion- C7*
  - *Wrist Extension- C6*
  - *Finger Flexion- C8*
  - *Finger abduction/Adduction C8-T1*
- <u>Check Peripheral Nerves</u>
  - *Ulnar, Radial, Median*
- Special tests
  - *Hoffman's, Spurlings, Lhermitte's*

www.aafp.org

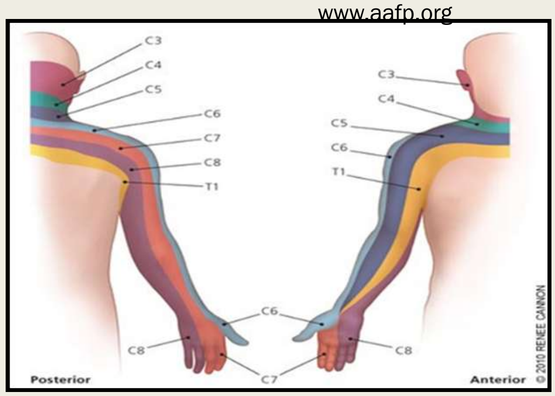
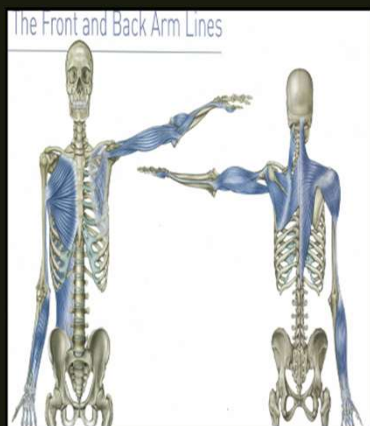


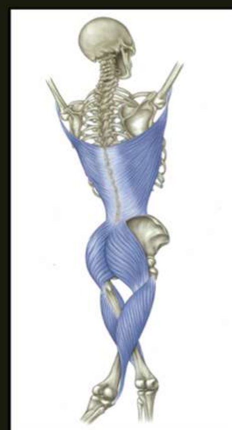
Table 1. Basic Nerve Functions and Exams[11,12]

NERVE	MOTOR	CLINICAL EXAM	SENSORY	CLINICAL EXAM
Radial	Wrist extensors	Extension of wrist, fingers, and thumb against resistance	Radial aspect of the dorsum of the hand, thumb, index finger, long finger, radial half of the ring finger proximal to the distal interphalangeal joints	Sensation at dorsal web space between thumb and index finger
Median	Muscles involving fine precision and pinch function of the hand, thenar muscles, index and long finger lumbricals	Opposition of the thumb to the fifth finger while watching the thenar muscles contract	Thumb, index, long, and radial side of the ring finger	Sensation at the volar tip of the index finger
Ulnar	Muscles involving grasping function, hypothenar muscles, interossei, adductor pollicis, ulnar lumbricals (two), deep head of flexor pollicis brevis	Abduction of fingers against resistance	Ulnar portion of dorsum of hand, fifth digit, and ulnar aspect of ring finger, hypothenar eminence	Sensation at the volar tip of the fifth digit

www.acep.org

APARAJITA SOHONI, M.D.





SHOULDER

Integrated Exam of the Glenohumeral, Scapulothoracic, Acromioclavicular and Sternoclavicular Joints

PICTURES FROM “Anatomy Trains” by Thomas W. Myers

Shoulder Evaluation

- **Observe** – posture, head, shoulders, thoracic spine (kyphosis? Scoliosis?), scapula, carrying angle
- **Inspect**- erythema, edema, displacement warmth, muscle atrophy, positioning at rest
- **Palpate**- SC, clavicle, AC, coracoid, head of humerous, bicipital groove, bicipital tendon, subacromial bursa, scapula, supraspinatus, infraspinatus, teres major and minor, subscapularis, latissimus, rhomboids, trapezius, deltoid, levator scapulae
- **Active & passive ROM, Strength Testing**- Internal Rotation, External Rotation, Abduction, Adduction, Flexion, Extension, circumduction, scapulothoracic ROM (protraction, retraction, depression, elevation, upward rotation) and feel for ROM and end feel- articular or soft tissue (joint below too)
- **Structural exam**, Mitchell Model, Zink Screen, upper crossed evaluation
- **Special tests**: Apley Scratch, Neer, Hawkins, Drop arm, empty can, glenohumeral traction-sulcus sign, speeds, scarf sign, Yearguson, apprehension, lift off test, Adson's
- **Somatic Dysfunction of the Shoulder:** Assess motion
  - *SC by springing (superior/inferior- shoulder shrug, anterior/posterior glide and anterior/posterior rotation),*
  - *springing of the AC AC inferior glide with Abduction, Superior glide with Adduction, AC posterior rotation with ER of the humerus and Anterior rotation with IR of the humerus,*
  - *Glenohumeral AP glide and superior/inferior glide*
- **Somatic Dysfunction**: tenderpoints, thorax, ribs, cspine, lspine, pelvis
- **Somatic Dysfunction**: affecting The A Autonomics, B Biomechanics, C Circulation, S Screening

Looks a lot like the COAR UE exam- but adding in somatic dysfunction assessment
Don't forget counterstrain points!

Shoulder Motion

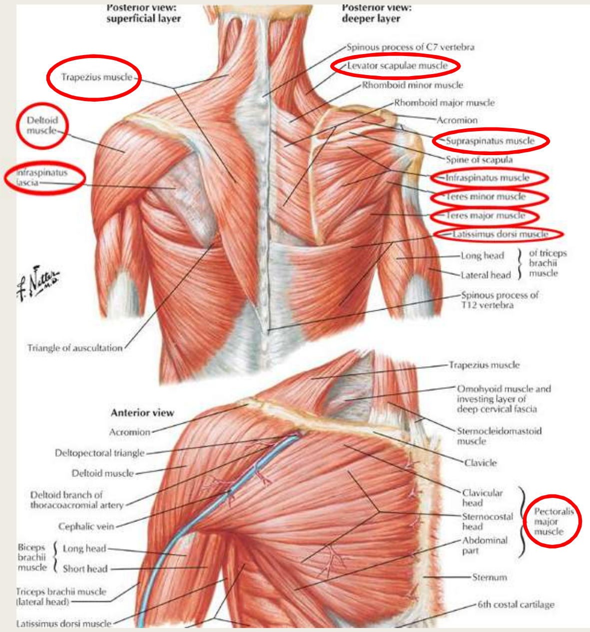
Elevation: upper trapezius, levator scapula

Abduction: deltoid (mostly middle), supraspinatus

Adducion: pectoralis major

Internal Rotation: subscapularis, pectoralis major, latissimus dorsi, teres major

External Rotation: infraspinatus, teres minor, (posterior deltoid assists ER)

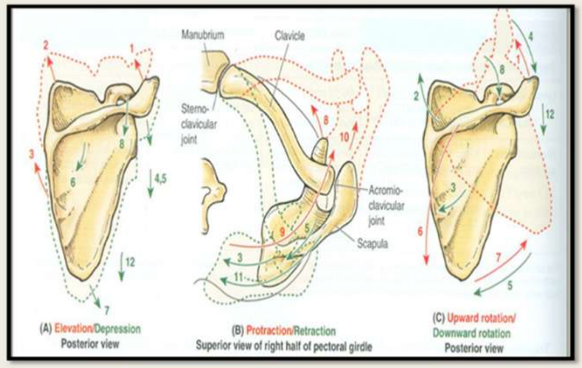


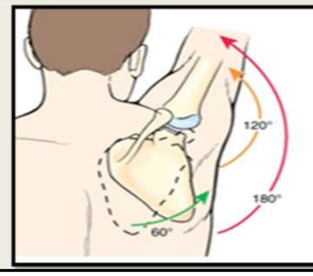
Elevate the scapula upper trapezius, levator scapula
Abduct the shoulder Deltoid, supraspinatus
Adduct the shoulder pectoralis major
IR the shoulder subscapularis, pectoralis major, latissimus dorsi, teres major
ER the shoulder infraspinatus, teres minor

SCAPULOTHORACIC JOINT

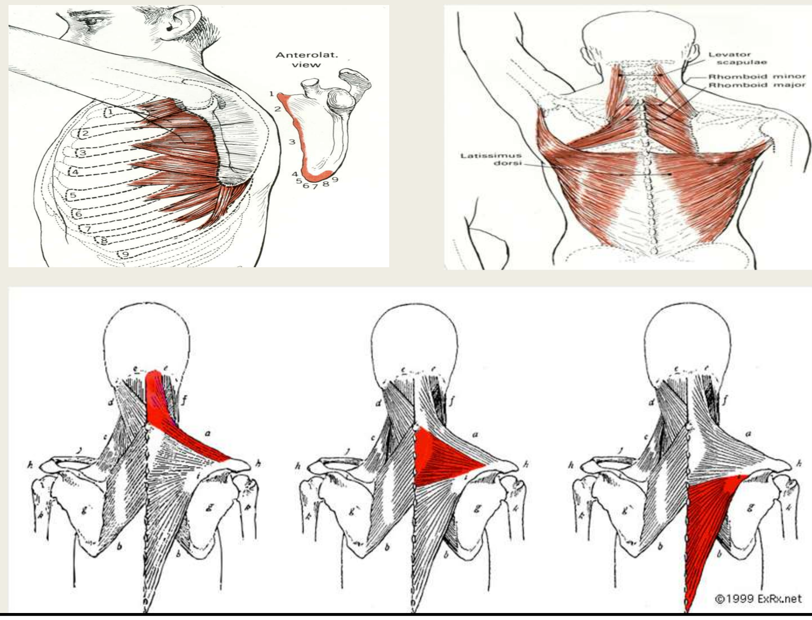
SCAPULOTHORACIC JOINT

- <u>MAJOR AND MINOR MOTIONS</u>
- <u>Scapulothoracic Joint & Active ROM</u>
- Protraction/Retraction
- *Assess by having the patient protract and retract (shoulder flexed to 90 degrees)*
- Elevation/Depression
- *Assess by watching shoulder shrug*
- Upward/Downward Rotation
- *Observe Shoulder Abduction*
  - *Normal Shoulder Abduction = 180 degrees*
  - *First 30 degrees glenohumeral has more motion*
  - *Then 4:1-7:1 ratio with glenohumeral moving more*
  - *Assess for Asymmetry (upper crossed- look for compensation by Levator Scapulae and upper trapezius)*
  - *Note restriction- which one moves sooner*
  - *Note early elevation by trapezius and levator scap*
- Thoracic, Rib and AC joint dysfunction can effect motion (as well as muscular/fascial dysfunction)





Shoulder Muscle Imbalance



Normal scapulothoracic motions that occur during arm elevation
Upward rotation
Posterior tilting
IR/EX

Due to sternoclavicular or acromioclavicular joints

Abnormal scapular motion is usually decreased posterior tilting, decreased upward rotation
Acromion may fail to move away from the humeral head during arm elevation
Reduction of subacromial space

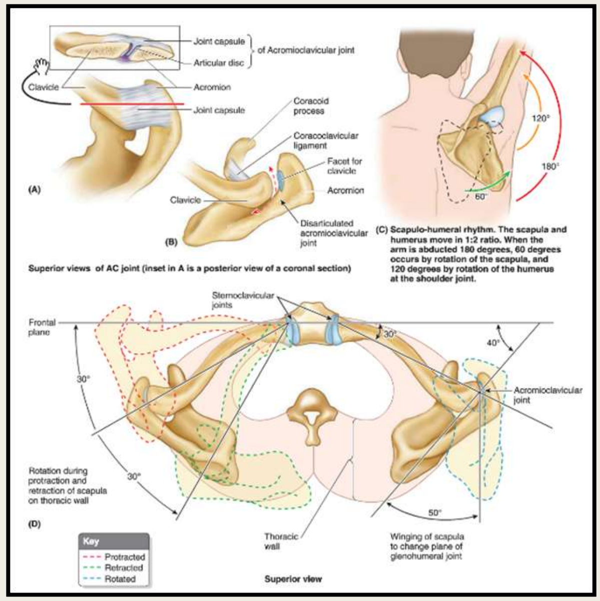
This could be due to tight pectoralis minor
Posterior shoulder tightness
Increased thoracic spine flexion or kyphosis

Serratus anterior and lower trapezius muscles stabilitze the scapula and induce upward rotation external rotation and posterior tilt allowing the humeral head to clear the acromia with elevation

STERNOCLAVICULAR JOINT

SC Joint – General Considerations

- True synovial joint
- Only true joint that attaches the UE to the thoracic cage
  - Saddle-like shape, irregular joint
  - Fibrocartilaginous articular disc (meniscus) divides joint into 2 compartments
  - Clavicle also articulates with rib 1
  - Motion critical to allow scapular motion (see picture)
- Little intrinsic stability *BUT* strong ligamentous attachment
  - Ligaments- prevent dislocation
- Anterior, posterior (prevent upward and lateral displacement)
- costoclavicular, interclavicular (prevent lateral displacement)
  - Bone will break before proximal clavicle dislocates
- Innervated by supraclavicular nerves C3,4
- Subclavius muscle
- Distal and proximal end moves in opposing motions



Subclavius draws the clavicle inferiorly and anteriorly
Origin- first rib and cartilage
Insertion subclavian groove of clavicle

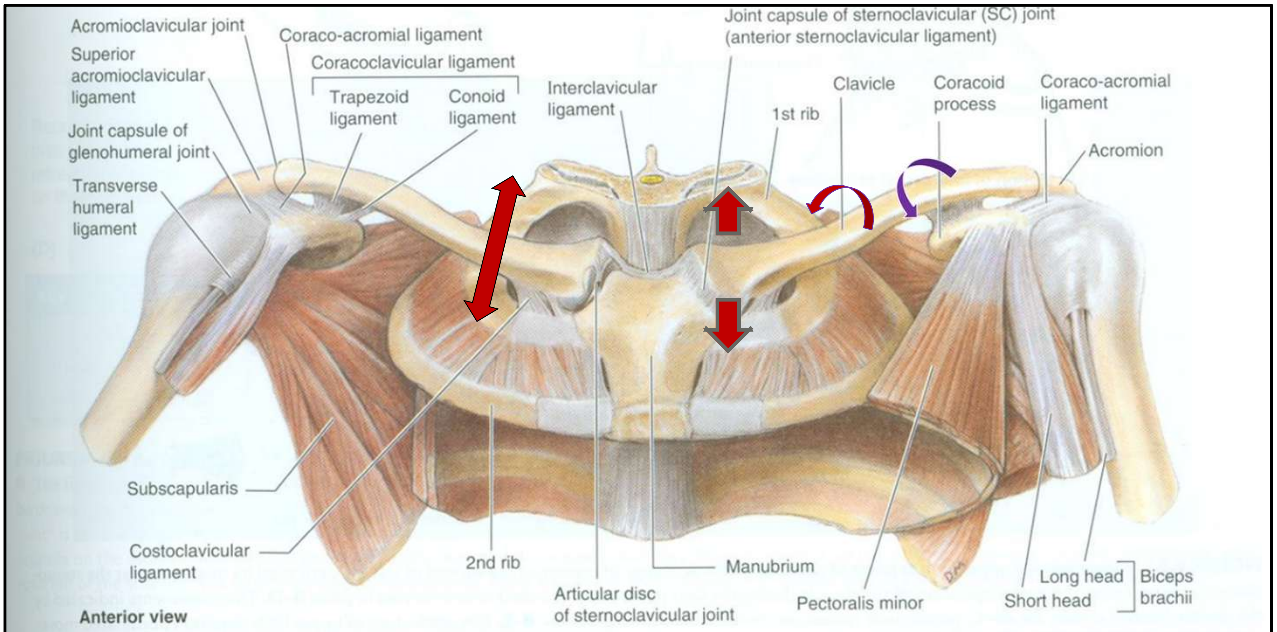


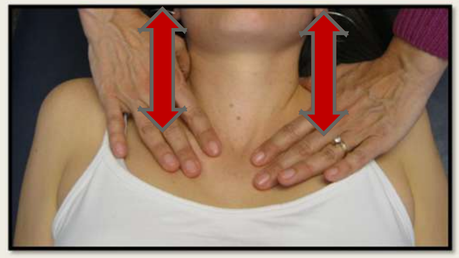
FIGURE 6.90. Pectoral girdle and associated tendons and ligaments. The pectoral girdle is a partial bony ring (incomplete posteriorly) formed by the manubrium of the sternum, the clavicle, and the scapulae. Joints associated with these bones are the sternoclavicular, acromioclavicular, and glenohumeral. The girdle provides for attachment of the superior appendicular skeleton to the axial skeleton and provides the mobile base from which the upper limb operates.

Consider the implications for scapular and shoulder motion if there is rib restriction?
How might this affect the thoracic inlet?
The first rib?
Shoulder abduction?
The brachial plexus?

- BEGIN BY SPRINGING!
- 3 sets of Motions permitted:

Sternoclavicular Joint

1. ***Superior–Inferior glides***
   - Tested with Shoulder shrug
   - As the distal clavicle elevates the proximal moves inferior
   - Superior shrug = inferior glide
   - Inferior shrug = superior glide
2. ***Anterior-Posterior glides***
   - Tested with protraction and retraction
   - As the distal clavicle moves anterior, the proximal should move posterior
   - Protraction = Posterior glide
   - Retraction = Anterior glide
3. ***Anterior-Posterior Rotation***
   - *Arm out to the side 90 degrees*
   - *Anterior rotation= IR*
   - *Posterior rotation = ER*



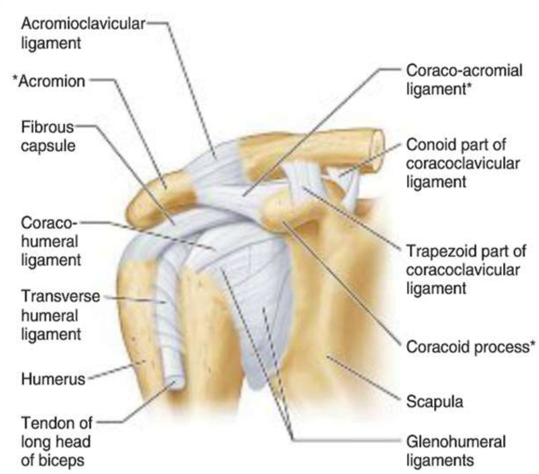
Only firm attachment of the upper extremity to the axial skeleton
Sometimes considered a “fulcrum” of the upper extremity

Consists of clavicle, first rib and a small facet on the inferomedial clavicle that articulates with the first rib

ACROMIOCLAVICULAR JOINT

AC Joint – General Considerations

- Two synovial cavities and an articular disc
- Ligaments
  - *Acromioclavicular*
  - *Coracoclavicular (trapezoid, conoid)*
- Motion
  - *Rotates on the acromion end of the scapula*
  - *Associated with motion of the scapulothoracic joint*
  - *No muscles move the AC joint*
  - *Scapular muscles cause the acromion to move on the clavicle*
  - *Importance of freedom of motion here for the scapula to move*
- Directly under the coracoacromian arch is the supraspinatus tendon and subacromial bursa



(B) Anterior view

* Coracoid process + coraco-acromial ligament + acromion = **coraco-acromial arch**

Discuss that the AC and SC do not have a muscle that directly attaches cross their joints to induce motion.

AC Joint Potential Motions:
anterior/posterior rotation & superior/inferior glide

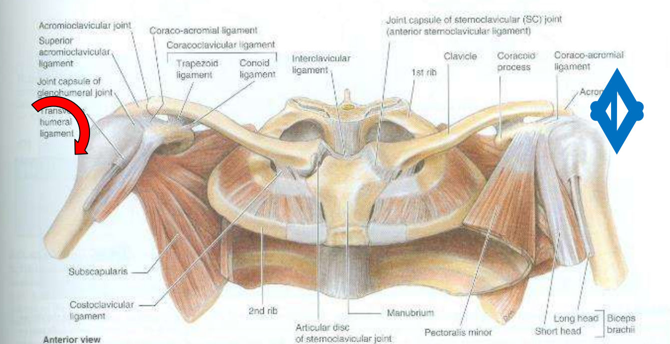
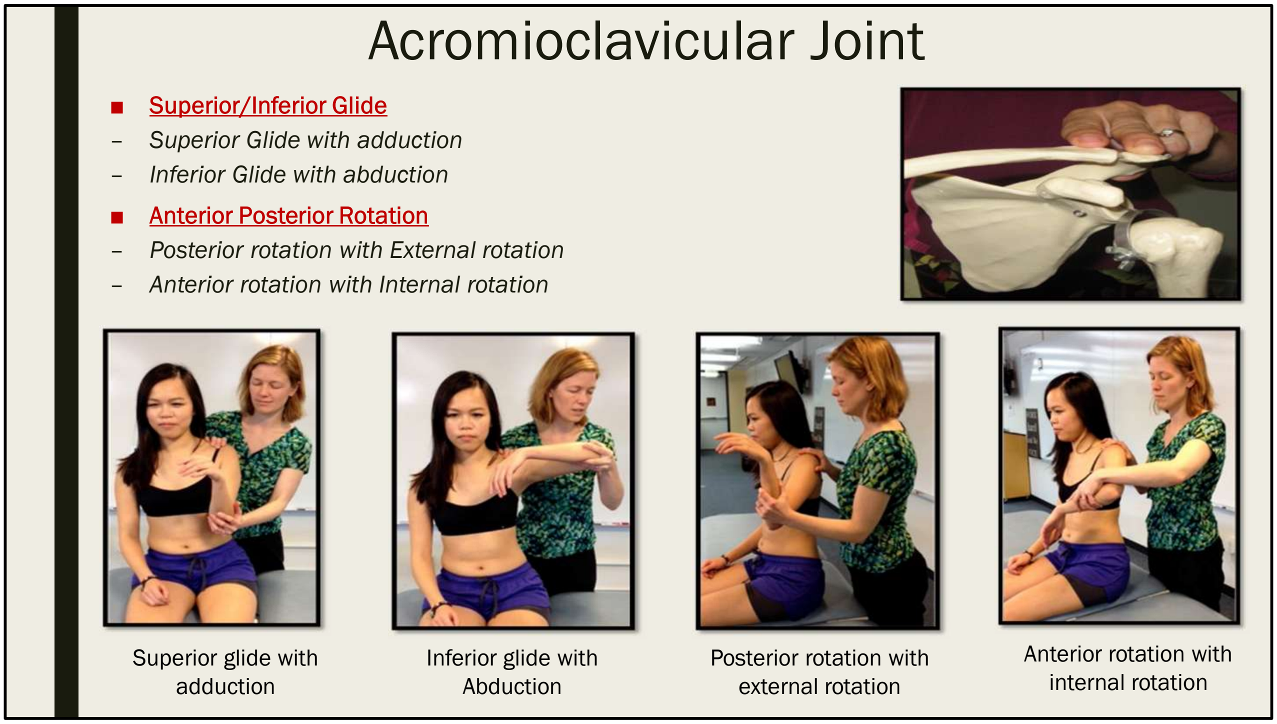


FIGURE 6.90. Pectoral girdle and associated tendons and ligaments. The pectoral girdle is a partial bony ring (incomplete posteriorly) formed by the manubrium of the sternum, the clavicle, and the scapulae. Joints associated with these bones are the sternoclavicular, acromioclavicular, and glenohumeral. The girdle provides for attachment of the superior appendicular skeleton to the axial skeleton and provides the mobile base from which the upper limb operates.

COA ,6th ed. p793

From Clinically Oriented Anatomy 6th Ed., page 793



Superior/Inferior Glide

Superior Glide of the acromion relative to the clavicle with Adduction of the humerus

Inferior Glide of the acromion relative to the clavicle with ABduction of the humerus

Anterior Posterior Rotation

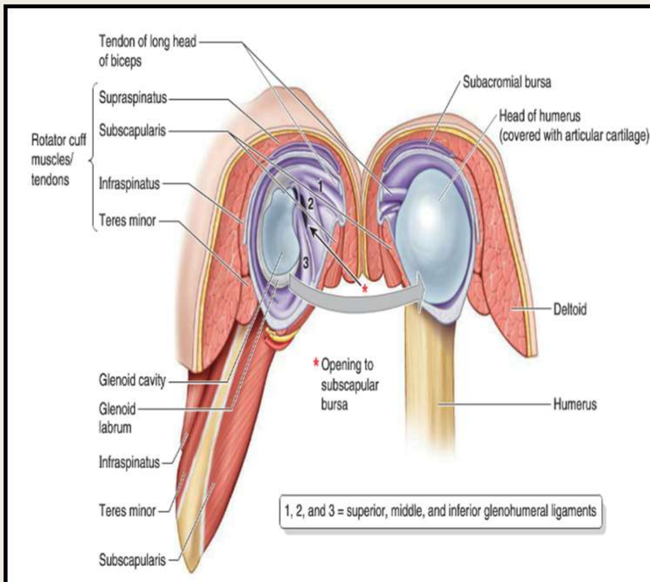
Posterior rotation of the acromion relative to the clavicle with External rotation of the humerus

Anterior rotation relative to the clavicle with Internal rotation of the humerus

GLENOHUMERAL JOINT

Glenohumeral Joint- General Considerations

- Head of the humerus as it articulates with the glenoid cavity of the scapula
- Labrum to help hold it in place
- Significant mobility but lacks stability
- Ligamentous joint capsule
  - *Posterior capsule commonly tight with causes anterior translation and elevation of the humeral head*
- Rotator cuff muscles really function to hold the humeral head on place during various motions



(A) Anterolateral view of glenoid cavity; posteromedial view of humerus

Glenohumeral Joint- Short Lever

Major Motions

- Flexion/Extension
- Abduction/Adduction
- Internal Rotation/External Rotation
- Circumduction

Minor motions-

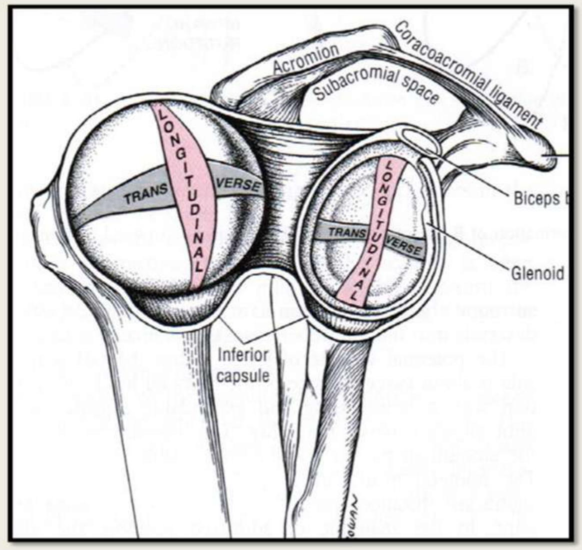
- Anterior /Posterior Glide
- Superior /Inferior glide
- Can Tx using Long or Short lever
- Stack eases/restrictions or achieve balanced tension

Glenohumeral Joint- Long Lever

Upper Extremity in Neutral Position

-seated or supine:

- Humeral Adduction = Superior glide
- Humeral Abduction = Inferior glide
- Humeral Internal rotation = Posterior glide
- Humeral External rotation = Anterior glide

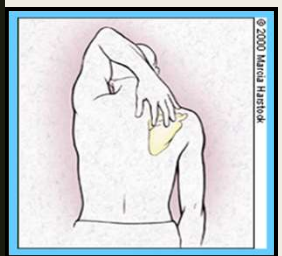


Summary of Shoulder Major and Minor Motions

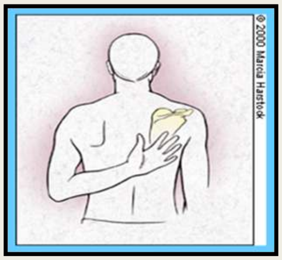
- <u>Scapulothoracic Joint</u>
  - Protraction/Retraction
  - Elevation/Depression
  - Upward/Downward Rotation
- <u>Acromioclavicular (AC) Joint</u>
  - Superior/Inferior glide
  - Anterior/Posterior Rotation
- <u>Sternoclavicular (SC) Joint</u>
  - Superior/Inferior Glide
  - Anterior/Posterior Glide
  - Anterior/Posterior Rotation
- <u>Glenohumeral Joint</u>
  - Internal/External Rotation
  - Abduction/Adduction
  - Flexion/Extension
  - Circumduction
  - Anterior/Posterior Glide
  - Superior/Inferior Glide

Shoulder Special Tests

Apley Scratch



Flexion, Abduction, ER



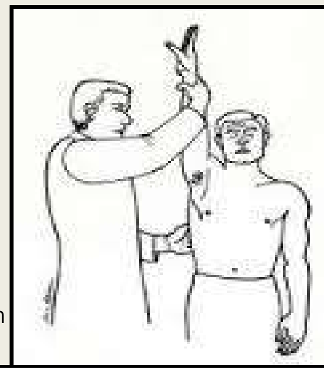
Extension, Adduction, IR

HAWKIN's-Kennedy

Internal rotation
Test is positive with pain in the subacromial space
Usually indicates impingement syndrome, supraspinatus dysfxn or Biceps dysfxn

Drop Arm Test

- Passively abduct patient's shoulder
- Observe as patient slowly lowers arm to waist
- If arm drops to patient's side, (unable to control descent) suggests rotator cuff tear and/or supraspinatus dysfunction
- Positive if there is pain- suggests supraspinatus tear or dysfunction



Neer

Passively flex the patient's arm while stabilizing the scapula- bringing the arm NEER to the EAR

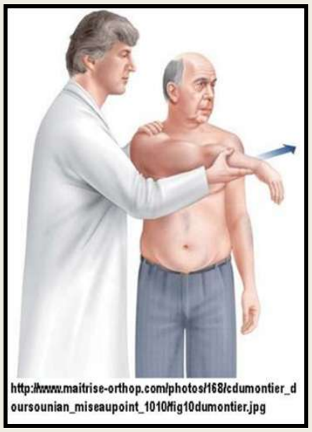
A positive test will elicit pain in the subacromial region

Special Tests

<u>AC Joint- Cross Chest Test/Scarf sign</u>

Adduction

Positive if pain localizes to the AC Joint



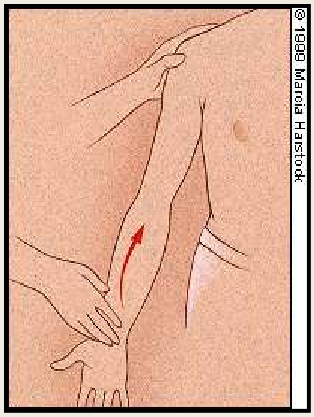
<u>Speed's Test</u>

Elbow flexed 20-30 degrees

Forearm supinated

Patient flexes and against examiner's resistance

+ test = pain in bicipital groove



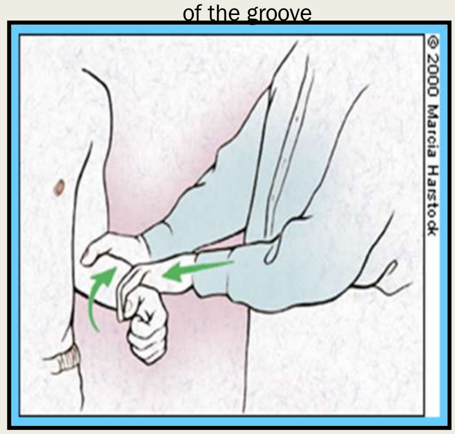
<u>Biceps Tendon Stability in Bicipital Groove/Yergason's Test</u>

Elbow flexed to 90 degrees

Grasp the patients hand (like a handshake)

Have the patient supinate against resistance

+ test = pain as tendon subluxes out of the groove



Special Tests

Glenohumeral Joint Traction
Sulcus sign
Pull inferiorly on the elbow
Observe for depression
Positive if > 1 cm
Indicates inferior instability

Shoulder apprehension
Place arm in 90 degrees abduction with elbow flexed
Apply pressure in a posterior direction
Caution if patient has a history of shoulder location
+ test if patient is apprehensive of dislocation

Lift Off Test-Subscapularis strength

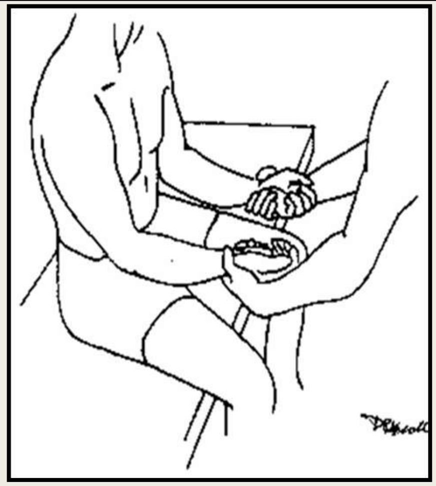
Adson's Test- feel for radial pulse- induce abd/ER/EX- if pulse disappears it is a positive test

THE ELBOW

Me: "My elbow hurts." WebMD: "Elbow cancer."

Elbow Evaluation

- *Observe* shoulder orientation, carrying angle
- *Inspect* warmth, edema, erythema, tenderness, deformity, alignment
- *Active ROM, Passive ROM, Muscle Strength Testing* (Flexion, Extension, pronation/supination)
- *Palpation*, Olecranon, radial head, medial/lateral epicondyle, medial/lateral collateral ligaments, wrist flexor attachment, wrist extensor attachment, Extensor wad, flexor wad, supinator, pronator, interossesous membrane
- *Special tests*
- *Resisted pronation, Resisted supination*
- *Resisted wrist flexion, Resisted wrist extension*
- *Assess Joint above and below* (Shoulder/Wrist)
- *Structural Exam,* Upper Crossed Syndrome, postural considerations
- *Somatic dysfunction of the area* -Tenderpoints, Trigger points, elbow Abduction/Adduction (medial/lateral glide), radial head motion (pronation/ supination)
- *Somatic dysfunction*
- Check for somatic dysfunction related to the ABC's



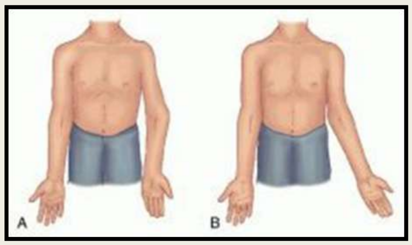
- Screen for somatic dysfunction of:
  - *Fascia*
  - *Interosseous membrane*
  - *Proximal radial head*
  - *Distal radioulnar joint*

Ulnohumeral Joint

- Hinge joint
- Surrounded by a capsule
- Muscles help to stabilize
- Held together in a spiral fashion:
  - *Extension- hand is away from the body*
  - *Flexion- hand is close to the body*
  - *(no motion required of humerus to do this)*
  - *Alignment/motion of the elbow significantly affects the wrist*

Significantly decreased carrying angle = gunstock deformity OR cubitus varus. Motion testing for SD = lateral glid ulna on humerus

Significantly increased carrying angle = cubitus valgus. Motion testing for SD = medial glid ulna on humerus



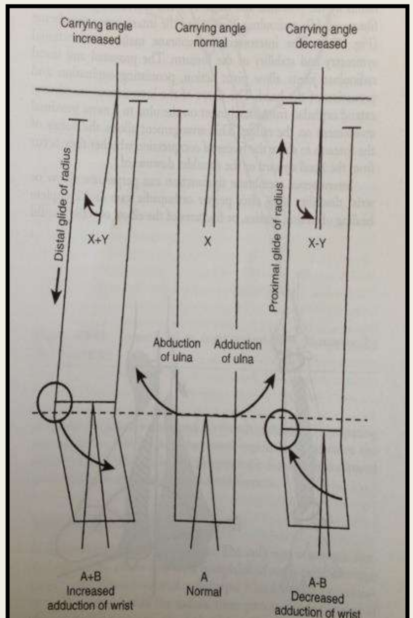


Figure 43-6 Carrying angle and parallelogram effect. (Illustration by W.A. Kuchera.)

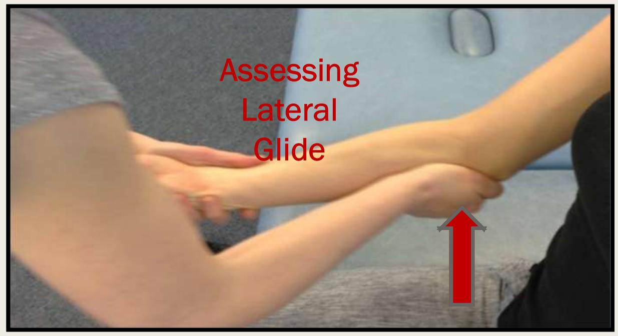
KNOW

Increased carrying angle- increased adduction of wrist

Decreased carrying angle- decreased adduction of wrist

Elbow Motions

- <u>3 Articulations</u>:
  - *Ulno-humeral (UH) Joint*
  - *Radio-humeral (RH) Joint*
  - *Proximal radio-ulnar (pRU) Joint*
- <u>Major</u>
  - *Flexion / Extension*
  - *Pronation/Supination*
- <u>Minor- used for Segmental Dx</u>
  - *Lateral glide proximally = ADduction distally*
  - *Medial glide proximally = ABduction distally*
  - *Pronation- posterior glide Radio-humeral/ulnar*
  - *Supination- Anterior glide Radio-humeral/ulnar*
- You can Diagnose the Radio-humeral, proximal radio-ulnar Joint & IOM
  Scan using Forearm Supination/Pronation (to Initial restriction to motion) to help lateralize to most restricted side

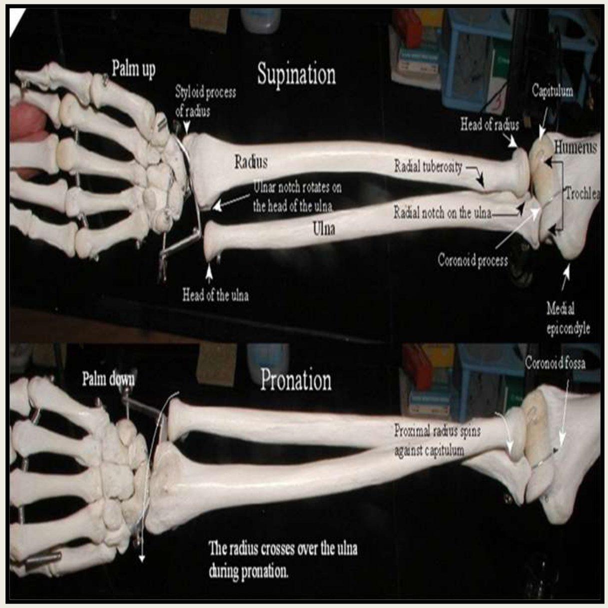


- -Thumbs on Radial styloid
- Pronate/supinate
- -Identifies potential SD of:
  - *-Forearm myofascia*
  - *-Interosseous membrane*
  - *-Radio-humeral (RH) Joint*
    -Proximal radio-ulnar (pRU) joint

RADIOULNAR JOINT

Radioulnar General Considerations

- Radial head articulation with the ulna
- Radial head rotates on the capitellum
- Elbow flexion- the radial head moves anteriorly on the capitellum
- Elbow extension- the radial head moves posteriorly on the capitellum
- Provides pronation and supination motions
- Hand Pronates → radial head glides Posteriorly
- Hand supinates → radial head glides anteriorly
- During ulnar abduction- radial head glides distally- wrist is pushed into adduction
- During ulnar adduction- radial head glides proximally- wrist becomes more abducted



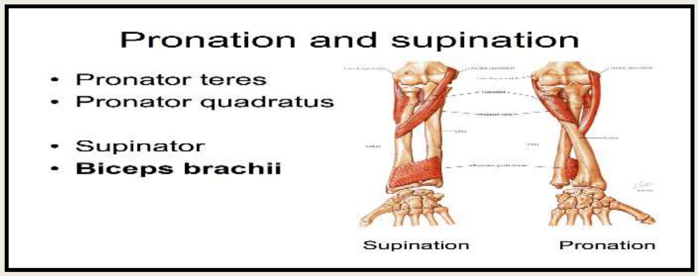
Radiohumeral Joint and IOM

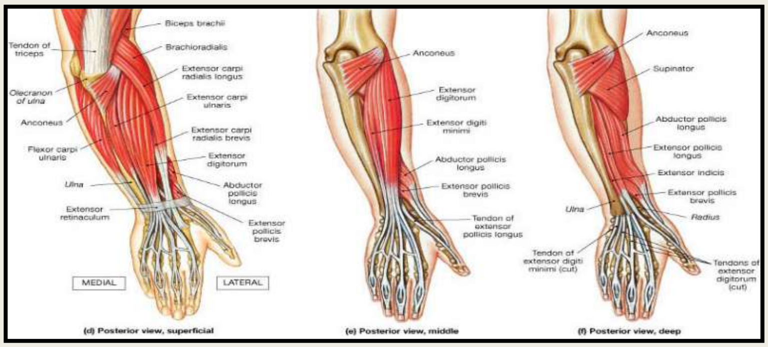
- Motion Test for restrictions to Anterior or Posterior glide of radial head
  - *Handshake grip*
  - *Posterior glide with pronation*
  - *Anterior glide with supination*
  - *Name for position of ease*
- Palpate Interosseous Membrane
  - *Along Dorsal surface for regions of Increased fascial tension & tenderness compared to other limb*
  - *Seg Dx = IOM restriction right or left*
  - *Interosseous membrane can perpetuate ongoing elbow or wrist disability even if the articular dysfunctions are corrected*

Medial/Lateral Epicondylitis

- Holding the patient's hand as in a handshake, resist supination and pronation.
- Then stabilize the wrist and try forced wrist flexion and extension
  - *Positive test is demonstrated by pain in either direction.*
  - *Pain with resisted supination or forced wrist extension is indicative of lateral epicondylitis (tennis elbow)*
  - *Pain with resisted pronation or forced wrist flexion is indicative of medial epicondylitis (golfer's elbow)*

- **Pronator**
  - *Pronator syndrome*
  - Compressive neuropathy of the Median nerve at the elbow
  - Can have paresthesias & forearm pain
- **Supinator**
  - *Posterior interossesous nerve syndrome- Radial nerve*
  - *Can have paresthesias & pain in the forearm*

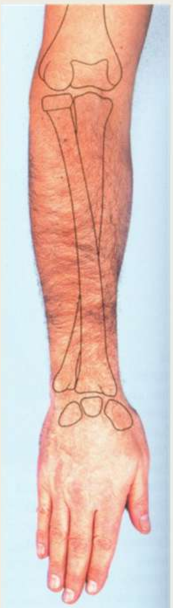


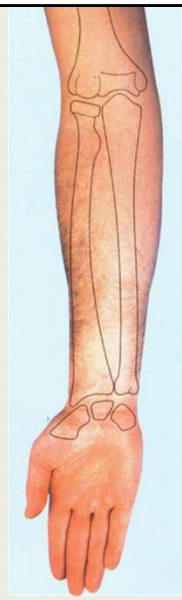


- **Wrist Flexors**
  - *Medial Epicondyle*
- **Wrist Extensors**
  - *Lateral Epicondyle*
- **Supinator**
  - *Origin- lateral epicondyle of humerus, supinator crest of ulna, radial collateral ligament, annular ligament*
  - *Insertion- lateral proximal radial shaft*
- **Pronator**
  - *Origin- medial supracondylar ridge of humerus/coronoid process of ulna*
  - *Insertion- lateral surface of the radius*

These muscles are all important and fair game for test material!!!!!!

Pronation





Supination

Fall on an Outstreched Hand

- *The radial head glides posteriorly in pronation and anterior in supination*
- *A BOARDS FAVORITE!!!*

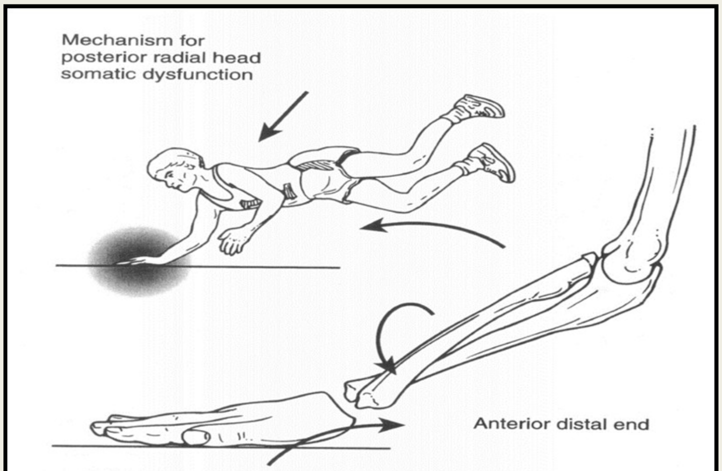
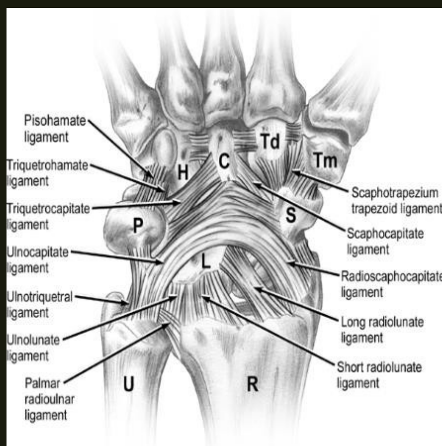


FIGURE 47.9. Forward fall on outstretched hand. (Illustration by W. A. Kuchera.)

WRISTS AND HANDS



Consider the vast interplay of any number of ligamentous and tendinous attachments- how could tension in the forearm cause restriction in the hand or wrist? How might bones in dorsal or plantar glide affect lymphatic or vascular structures?

Integrated Evaluation of the Wrist/Hand

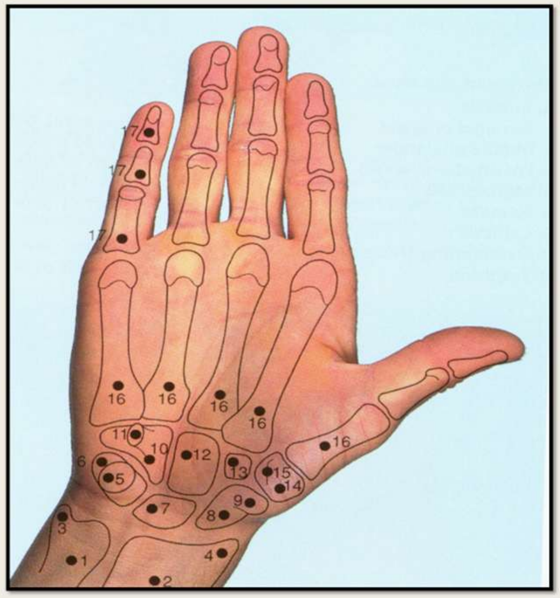
- ***Observe*** *posture, shoulder position, carrying angle*
- ***Inspect*** *for warmth, edema, erythema, tenderness, Joint deformities, joint positions*
- ***Active ROM, Passive ROM , Muscle Strength Testing****(flexion, extension, supination, pronation, radial deviation, ulnar deviation, circumduction, flex and extend fingers, abduct and adduct fingers, thumb opposition*
- ***Vasculature/lymph nodes:*** *Radial/Ulnar artery- Allen's test, epitrochlear nodes, axillary nodes*
- ***Innervation:*** *strength, sensation, reflexes*
- ***Palpation****, interosseous membrane, distal radius and ulna, scaphoid, lunate, triquetrum, pisiform, trapezoid, trapezium, capitate, hamate, metacarpals, phalanges, carpal tunnel, wrist and finger flexors and extensors, thenar/hypothenar eminence*
- ***Special tests***
- *Palpation of snuff Box*
- *Phalen/Reverse phalen*
- *Tinel- carpal tunnel, guyon's canal, cubital tunnel*
- *Finkelsteins*
- ***Assess Joint above and below*** *(Fingers, Elbow)*
- ***Structural Exam,*** *Upper Crossed Syndrome, Mitchell Model, Zink Screen*
- ***Somatic dysfunction of the area*** *–Tenderpoints, wrist minor motions in flexion/extension, radial/ ulnar deviation, distal radioulnar A/P glide, springing of carpals (looking for restriction or laxity), minor motions of the wrist and hand*
- ***Somatic dysfunction*** *affecting the A Autonomics, B Biomechanics, C Circulation S Screening*

Radiocarpal & Radioulnar Joints

Major Motions:

- *Flexion/extension*
- *AB/ADduction*
- *Circumduction*
- *Named in Anatomic Position*

- Radiocarpal joint:
  - *Radius articulating with Scaphoid, Lunate & Triquetrum bones*
- Seg Dx= Ab/Adduction and Flex/ extension somatic dysfunction
- Distal radioulnar joint:
- Seg Dx = Anterior or posterior glide somatic dysfunction



Carpal/Metacarpal/Phalangeal Joint Motion

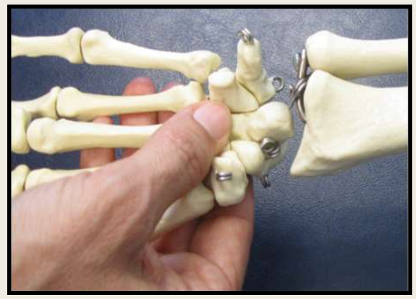
Inter-carpal joints (IC)= Palmar/ Dorsal glide SD

Carpometacarpal joints (CMC) = Palmar/ Dorsal glide (except thumb)

Metacarpo-phalangeal joints (MCP's) and Proximal and distal interphalangeal joints (PIP's DIP's) all have same motions: With respect to midline of body

- *Ant (Palmar)/ Post (Dorsal) glide*
- *Medial/ lateral glide*
- *Internal/ External rotation*

- You can SPRING the carpals for rapid identification of restricted bones



Carpal Tunnel Tests

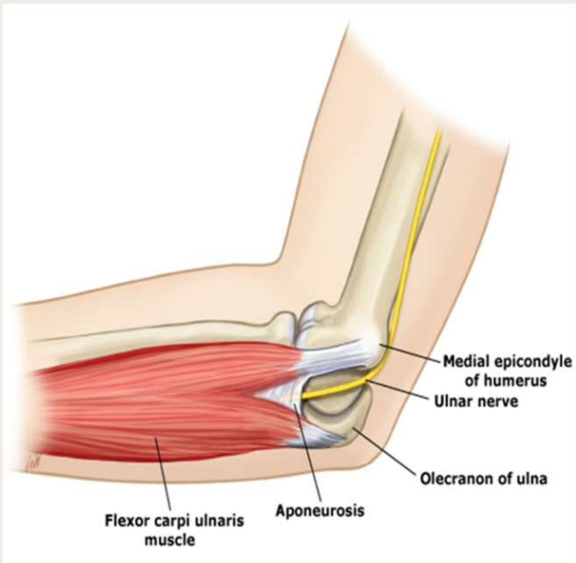
Positive Phalen's, Reverse Phalen's and Tinel's will recreate paresthesias in the first three digits

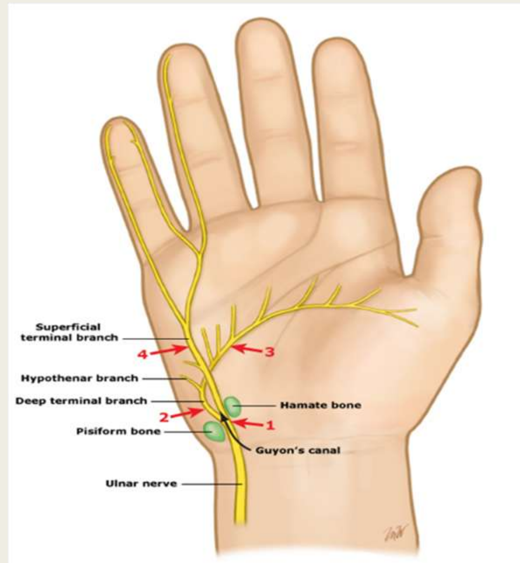
Phalen's & Reverse Phalen's

Tinel's Test

Thenar and Hypothenar atrophy

Ulnar Nerve Compression- Guyon Canal and Cubital Tunnel Syndrome





Know where the ulnar nerve can get hung up

Putting It all Together- practice doing these ROM with YOUR Palpation and Gross ROM tests!!

THE SHOULDER

MAJOR AND MINOR MOTIONS

Scapulothoracic Joint

- Protraction/Retraction
- Elevation/Depression
- Upward/Downward Rotation

Acromioclavicular (AC) Joint

Superior/Inferior glide

Anterior/Posterior Rotation

Sternoclavicular (SC) Joint

Superior/Inferior Glide
Anterior/Posterior Glide
Anterior/Posterior Rotation

Glenohumeral Joint

Internal/External Rotation
Abduction/Adduction
Flexion/Extension
Circumduction
Anterior/Posterior Glide
Superior/Inferior Glide

THE ELBOW

Ulnohumeral Joint

Flexion/Extension
Abduction/Adduction

Radiohumeral

Anterior/Posterior glide

THE WRIST

Radiocarpal

Flexion/Extension
Radial/Ulnar deviation
Circumduction

Distal Radioulnar

Anterior/Posterior Glide

THE HAND

Metacarpals/phalanges

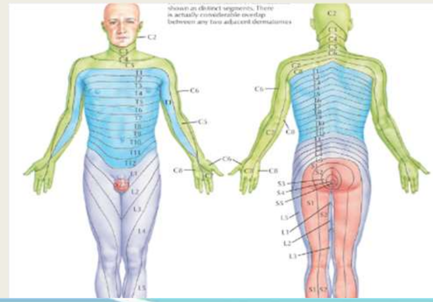
Flexion/Extension
Abduction/Adduction
IR/ER, Dorsal/Palmar, Medial/Lateral

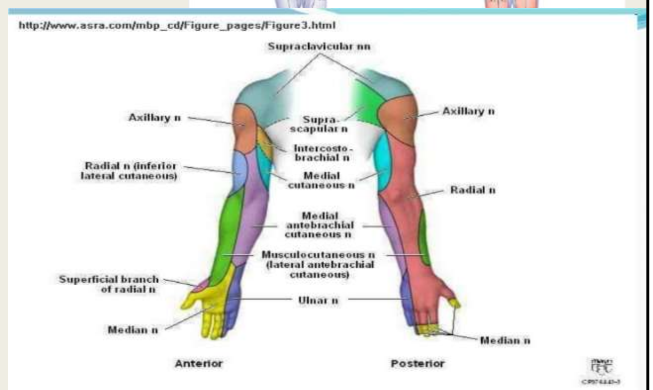
I forgot to put carpals on here- they all have A/P Glide

FINAL THOUGHTS

Consider REFERRAL Pain

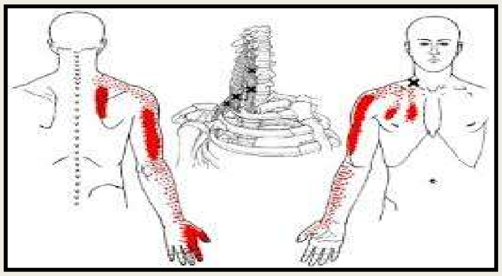
- Dermatomal Referral Patterns
- Neurologic exam
  - *Evidence of deficit?*
  - *Consider performing before MSK exam*
- History & mechanism of injury
  - *Suggestive of spine/disc insult?*
  - *Age/PMHx/Degenerative change?*
  - *Is described pain dermatomal?*
  - *Rheumatologic signs/symptoms*
- Provocative evaluation
  - *Special tests - Spine*
  - *Local tenderness to palpation*
- Sclerotomal Referral Patterns
- Myotomal Referral Patterns
- Consider non- MSK causes- Referral from Heart/Lungs/Gallbladder/Diaphragm etc

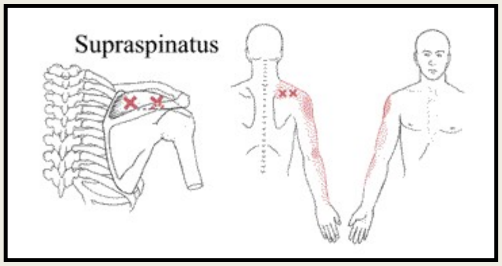


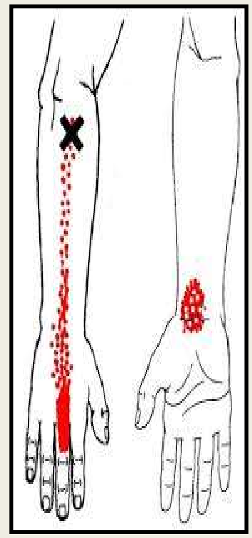


Trigger Points/Enthesopathy

- Trigger points in particular locations can cause radiating pain in distant locations
  - Can be treated with direct inhibition, spray n stretch or injections
- Ligamentous laxity, enthesopathy can also cause pain in a sclerotomal referral pattern
  - Check at attachment points of tendons and ligaments
  - Prolotherapy may be beneficial







How OMT can help patients with UE complaints

- OMT can affects circulation and lymphatic flow via removal of mechanical obstruction and normalization of autonomic tone
  - *Stasis promotes abnormal intra/extracellular environments*
  - *Can lead to fibrosis and adhesions*
  - *Removal of edema from injury*
  - *Ability to heal through proper circulation- tissue perfusion*

- SOMATIC DYSFUNCTION
  - Can be a primary cause of pain, dysfunction, and complaints of the upper extremity
  - *LYMPHATICS/CIRCULATION*
  - *Promote movement of fluids (Thoracic inlet, Diaphragm, carpal tunnel, antecubital fossa, axilla)*
  - *BIOMECHANICAL*
  - *Look for local, regional and inter-regional biomechanical issues, somatic dysfunction that may affect the complaint (upper crossed syndrome, posture, carrying angle, ergonomics)*
  - *NERVOUS SYSTEM*
  - Address the sympathetic/parasympathetic innervation to the area- treat related structures for autonomic balance (sympathetics T2-T7)
  - Tropic function of neuronal axons impeded by mechanical compression
  - NOCICEPTION
  - Alleviate pain at site, reduce neuronal facilitation, reduce central sensitization, decrease allostatic load

OMT sets the stage for healing overuse injuries

Sheldonosteopathicclinic.co.uk

- Osteopathic Approach to the Patient
  - *Decrease Allostatic Load*
  - *Optimize Homeostatic Mechanisms*
  - *Promote Self healing*

- USE OMT TO:
- Alleviates biomechanical issues which would inhibit proper exercise/rehabilitation
- Decreases local swelling
- Improves pain tolerance
- Improves innate healing power via fluid transfer and improved perfusion
- Decreases need for pain medication
- Improves ability to compensate for injury

Contraindications to Extremity OMT & BLT/MFR

- Over direct fracture site
- Over direct infection site
- Directly over a tumor
- Acute emergent situations involving the UE
- Complete ligamentous/tendinous rupture
  - *(2nd or 3rd degree sprains, active tissue bleeding)*
- Lack of Somatic dysfunction or unsure of diagnosis
- Suspected urgent Visceral/Neural disease
- HVLA contraindicated with extreme laxity- ie sulcus sign

- OMT is generally safe and effective even for acute symptoms/syndromes
  - *Especially if using gentle techniques like BLT/MFR, CS, soft tissue, lymphatics*
- Even given Contraindications above can often direct treatment at a different region to affect, sympathetics, lymphatics, vascular, biomechanics and help restore homeostasis
- Absolute- lack of somatic dysfunction, lack of patient/preceptor consent, lack of patient cooperation

Balanced Ligamentous Tension & Myofascial Release

- Great to treat the Upper extremity
- Good for acute injuries as well as chronic somatic dysfunction
- The better you know your anatomy- the better you will be at balancing or creating a release
  - *Vectors, tension, connections*
- See lab handout AND Lab Powerpoint for description of techniques
- <u>BLT</u>
  - *Take tissues to a point of balanced tension*
  - *Hold there until release happens*
  - *No following tissues- the release happens around what you are holding*
- <u>MFR</u>
  - *Take tissues to point of maximal ease (indirect) or into the barrier (direct)*
  - *Hold until a release*
  - *Can follow the unwinding*
  - *May use respiration to help facilitate a release*

Bibliography

- Foundations of Osteopathic Medicine 3rd Edition
  - *Chapter 43 Upper Extremities*
- Moore's Clinical Anatomy
- Thieme Atlas of Anatomy 3rd edition
- COAR Upper Extremity Packet